C3300-48FS

Specification

Hardware	Description	
Performance	• Switching Capacity: 216 Gbps	• Forwarding rate: 160.7 Mpps
Interface	• 48 port 100/1000 Base-X	• 4 port 1G/10G
	• 1 Extension slot: C3100-NM-2X(2 port 1G/10G)	
Memory	• DRAM 2 GB	• Flash 512 MB / eMMC 8GB
FAN	• Dual Fixed FANs	
Management	• 1-port RJ-45 Console	• 1-port USB
Power Supply	• Dual power supplies	• Modular power supply with hot-swappable
	• Power Consumption: 76 W	• Heat Dissipation: 259.4 BTU/h
Environment	• Operating Temperature: 0~50℃	• Operating Humidity: 10~90% (Non-condensing)
Dimension	• 440(W) x 420(D) x 44(H) mm	• 1 RU

Software	Description	
Virtualization	• VS: Virtual Stacking	• Max 8 Units Stack
	• MAD: Multi-Active Detection	• Multi VRF
Entry	• 32K MAC Table	• 12K IPv4 routes
	• 4K Active VLAN	• 12KB Jumbo frame
VLAN	• MAC / Protocol / IP Subnet based	• QinQ
	• 802.1Q	• Private VLAN / Super-VLAN
	• Voice VLAN	• GVRP
STP / Ring Protocol and LP	• Standard STP / RSTP / MSTP	• PVSTP+ Compatible protocol
	• Loop Detection and Prevention	• ERPS
	• BPDU Guard / BPDU Filter / Root Guard	• Link Flap Detection
Aggregation	• Static Link Aggregation	• Dynamic Link Aggregation(LACP)
	• Member per group: 8	

Software	Description	
Multicast	• IGMP Snooping(with Querier and Fast Leave)	• IGMP v1/v2/v3
	• IGMP Filter	• PIM-DM/SM/SSM
IPv6	• IPv4/IPv6 dual stack	• IPv6 Management
	• IPv6 ACL	• MLD Snooping
	• VRRPv3	• RIPng / OSPFv3 / BGP4+
L3 Features	• Support full L3 features without additional license	
	• Static / RIP / OSPF / ISIS / BGP	• PBR(Policy-based Routing)
	• BFD	• IPSec encryption for routing protocol
	• VRRP	• Gateway Keepalive
Security	• L2~L4-based ACL	• Port ACL, VLAN ACL, Global ACL
	• Time-based ACL	• 802.1x authentication
	• Port Security	• DHCP Snooping
	• Dynamic ARP Inspection	• IP Source Guard
	• DoS/DDoS Protection	• Anti-ARP Spoofing prevention
	• CPP(CPU Protection Policy)	• MAC Address Limiting
	• Enforced Password Change Upon First Login	• Strong Password Policy and Encryption Mechanism
	• ID/PW authentication *	• User isolation after APT prevention system interoperability *
	• MAC based authentication *	• Checking the required security configuration *
	• Network segmentation *	• User control by schedule *
QoS	• 8 Queues	• Queue Scheduling: SP / WRR / SP+WRR
	• Classification and Marking with DSCP /802.1p	• Traffic Shaping and Policing
	• Policy-Based QoS	• Tail-drop and WRED(Weighted Random Early Detection)
	• Buffer Management	• Upload / download bandwidth control per user *
SDN	• Netconf	
Reliability	• OS Image Redundancy	• Configuration Redundancy
	• Hot Patch	• Hardware Watchdog
Management Features	• Ping and Traceroute	• Console, Telnet and SSHv2
	• SNMPv1/2c/3 and RMON	• LLDP / LLDP-MED
	• Mirroring / Remote mirroring / VLAN mirroring	• 1:1, N:N Mirroring

* CoreEdge Smart Controller required.

Software	Description	
Management Features	• NTP	• DHCP Server/Relay/Client
	• AAA(Authentication, Authorization, Accounting)	• RADIUS & TACACS+
	• Syslog	• Configuration Rollback
	• DDM(Digital Diagnostic Monitoring)	• ULFD (Uni-Directional Link Fault Detection)
	• Flow Monitoring	• RBAC(Role-Based Access Control)
	• RTR(Response Time Reporter) for quality checking	• BSM features for switch diagnostics
	• Embedded Event Script	• FTP / SFTP
	• CSS is a sophisticated suite of management tools that provides a simplified approach.	
	• ZTP(Zero Touch Provisioning) *	• Sanity Report for switch and terminal *
	• Scheduled firmware upgrade *	• Scheduled config backup *
	• Auto-Backup/Upgrade/Provisioning/Recovery *	• End Device Management *
	• Switch Shaped Panel *	• System Health Monitoring *
	• IP & Asset Management *	• Terminal Vulnerability Scanning *
	• Dynamic Policy Enforcement *	
Test Report	• Performance Test Report – Electronics and Telecommunications Research Institute(ETRI)	
Certification	• KC Electro-Magnetic Compatibility – National Radio Research Agency(NRRA)	
	• Security Functionality Certificate – National Intelligence Service(NIS)	

* CoreEdge Smart Controller required.

Standards	Description	
IEEE	• IEEE 802.3ab 1000BASE-T	• IEEE 802.3z 1000BASE-X
	• IEEE 802.3ae 10G BASE-X	• IEEE 802.3an 10G BASE-T
	• IEEE 802.3az EEE(Energy Efficient Ethernet)	• IEEE 802.3x Flow Control
	• IEEE 802.3ad Link Aggregation Control Protocol	• IEEE 802.1AX Link Aggregation
	• IEEE 802.3ah (Ethernet in the First Mile)	• IEEE 802.1ag CFM(Connectivity Fault Management)
	• IEEE 802.1D STP(Spanning Tree Protocol)	• IEEE 802.1w RSTP(Rapid Spanning Tree Protocol)
	• IEEE 802.1s MSTP(Multiple Spanning Tree Protocol)	• IEEE 802.1x(Port-Based Network Access Control)
	• IEEE 802.1ad Provider Bridges(Q-in-Q)	• IEEE 802.1Q VLAN Tagging
	• IEEE 802.1v VLAN Classification by Protocol & Port	• IEEE 802.1AB LLDP(Link Layer Discovery Protocol)

Standards	Description	
RFC	• RFC 768 UDP (User Datagram Protocol)	• RFC 791 IP (Internet Protocol)
	• RFC 792 ICMP (Internet Control Message Protocol)	• RFC 793 TCP (Transmission Control Protocol)
	• RFC 826 ARP (Address Resolution Protocol)	• RFC 854/855 Telnet and Tenet options
	• RFC 894 Core IP Implementation for Ethernet	• RFC 919 IP Subnet Broadcasting
	• RFC 950 & RFC 951 Internet Standard Subnetting Procedure & BootP	• RFC 1027 Proxy ARP
	• RFC 1035 DNS Client	• RFC 1042 IP over IEEE 802 Networks
	• RFC 1058 RIP (Routing Information Protocol)	• RFC 1071 Internet Checksum Calculation
	• RFC 1122 Internet Host Requirements	• RFC 1157 SNMP
	• RFC 1191 Path MTU Discovery	• RFC 1213 MIB-II
	• RFC 1239 Standard MIB	• RFC 1350 TFTP (Trivial File Transfer Protocol)
	• RFC 1591 DNS (Domain Name System)	• RFC 1724 RIPv2 MIB Extension
	• RFC 1812 Requirements for IPv4 Routers	• RFC 1918 IP Addressing
	• RFC 1981 Path MTU Discovery for IPv6	• RFC 2082 RIP-2 MD5 Authentication
	• RFC 2131 DHCPv4 Client	• RFC 2328 OSPF Version 2
	• RFC 2453 RIPv2	• RFC 2460 IPv6 Specification
	• RFC 2464 Transmission of IPv6 Packets over Ethernet Networks	• RFC 2578 SMIv2
	• RFC 2581 TCP Congestion Control	• RFC 2711 IPv6 Router Alert Option
	• RFC 2819 RMON MIB (groups 1,2,3 & 9)	• RFC 2863 Interfaces Group MIB
	• RFC 2865 RADIUS Authentication	• RFC 2866 RADIUS Accounting
	• RFC 3021 Using 31-Bit Prefixes on IPv4 Point-to-Point Links	• RFC 3046 DHCP Relay Agent Information Option
	• RFC 3315 DHCPv6 Client	• RFC 3411 SNMP Architecture
	• RFC 3414 USM for SNMPv3	• RFC 3416 v2 of the Protocol Operations for the SNMP
	• RFC 3418 MIB for SNMP	• RFC 3587 IPv6 Global Unicast Address Format
	• RFC 3484 Default Address Selection for IPv6	• RFC 3596 DNS Extensions for IPv6
	• RFC 3646 DNS Configuration Options for DHCPv6	• RFC 3768 VRRP
	• RFC 4007 IPv6 Scoped Address Architecture	• RFC 4022 TCP (MIB for the Transmission Control Protocol)
	• RFC 4113 MIB for the UDP (User Datagram Protocol)	• RFC 4213 Transition Mechanisms for IPv6 Hosts & Routers
	• RFC 4291 IPv6 Addressing Architecture	• RFC 4292 IP Forwarding Table MIB
	• RFC 4293 MIB for the IP (Internet Protocol)	• RFC 4318 Definitions of Managed Objects for Bridges with RSTP
	• RFC 4330 SNTP version 4	• RFC 4443 ICMPv6
	• RFC 4861 Neighbor Discovery for IPv6	• RFC 5014 IPv6 Socket API for Source Address Selection
	• RFC 5095 Deprecation of Type 0 Routing Headers in IPv6	• RFC 5424 Syslog Protocol
	• RFC 5905 NTP version 4	• RFC 4607 Source-Specific Multicast for IP
	• RFC 3973 PIM-DM	• RFC 7761 PIM-SM